PROGRESS CHECK

People Counting with YOLO11 and ByteTrack | p29 | 26 October 2026

Current state: the end-to-end pipeline is working. Four single-detector configurations and a two-detector ensemble have been evaluated. All numbers below are preliminary comparisons, not an independent final test.

1. Completed experiments

- Data: MOT17-04/05 yielded 1,510 training images (39,144 boxes) and 377 chronological validation images (11,397 boxes). Four other videos from MOT17/MOT20 provided 2,454 comparison frames.
- Approach 1: COCO YOLO11n and YOLO11s, plus YOLO11s fine-tuned for 3 or 20 epochs, were run with the same custom ByteTrack and directional line counter. The 3-epoch model led the single-detector CA comparison.
- Approach 2: the 3-epoch model and COCO YOLO11s run together; their person boxes are fused at cross-model IoU 0.55 before the unchanged tracker and counter. No extra detector training was required.
- The repository includes data/training/inference/evaluation scripts, the selected checkpoint, recorded JSON results, and 42 passing tests.

2. Preliminary comparison

Mean values weight each of the four comparison sequences equally. Higher CA, MOTA, and IDF1 are better; lower cumulative-count curve MAE is better. CA reflects errors in final IN/OUT totals.

Configuration	CA	Curve MAE	MOTA	IDF1
COCO YOLO11n	0.5501	2.5913	0.4361	0.4446
COCO YOLO11s	0.6222	1.9747	0.4352	0.4671
YOLO11s, 3 epochs	0.6548	1.7021	0.4113	0.4324
YOLO11s, 20 epochs	0.5794	2.1542	0.4080	0.4277
Two-detector ensemble	0.6654	1.7858	0.3989	0.4261

Scene-level counting

Comparison video	3-epoch CA	Ensemble CA	Change
MOT17-02	0.5238	0.5714	+0.0476
MOT20-01	0.3462	0.5385	+0.1923
MOT17-09	0.9231	0.7692	-0.1539
MOT17-11	0.8261	0.7826	-0.0435

The ensemble's mean CA is 0.6654 versus 0.6548 for the 3-epoch single model (+0.0106). Its largest gain is on crowded MOT20-01. It takes 164.7 s versus 125.4 s for the same 2,454 frames (about 31% more time).

3. Observations and difficulties

- The ensemble cuts tracker misses from 26,801 to 23,491, but raises false positives from 5,812 to 9,076. The trade-off helps dense-scene counting while mean MOTA and IDF1 decline.
- Its final-count CA rises, but curve MAE worsens from 1.7021 to 1.7858. A closer final total does not imply better counts throughout the video. CA also drops on MOT17-09 and MOT17-11.
- Twenty epochs raise validation mAP50-95 from 0.6781 to 0.7246, yet comparison CA falls from 0.6548 to 0.5794. The validation frames share the two training cameras, so new-view performance cannot be inferred from that mAP alone.
- Tracking has no appearance/ReID branch; long occlusions can break identities. Counting-line geometry is set manually per camera. A VisDrone aerial experiment remains unavailable.
- An OpenCV warpAffine failure interrupted the 20-epoch run with Windows data-loader workers. It resumed successfully with workers=0.

Protocol limitation: the four comparison videos were excluded from detector training, but their scores informed model choice. They are exploratory post-selection comparisons. The current 20% validation split also shares cameras with training.

4. Remaining work

- Add camera-disjoint development data and use it to select fusion, confidence, and tracker settings without reusing the comparison set.
- Review missed and extra IN/OUT events near each line; connect them to missed detections, false positives, identity switches, and direction errors.
- Test track-lifetime/association changes and, if feasible, appearance features; measure throughput and memory consistently.
- Obtain an aerial sequence and evaluate the existing polygon entry/exit mode with documented camera geometry.
- Freeze the full pipeline, then evaluate once on untouched cameras. Report CA, curve MAE, MOTA, IDF1, speed, and failure cases.

Repository and evidence

https://github.com/NikolayTaran/cv_people_counting | Source: docs/PROGRESS_CHECK.md and machine-readable results in reports/. The code names the comparison split 'test'; it is not an untouched final test.